

Enzo Porto Brasil

Statistical Inference, Uncertainty & Scientific Computing

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I am a statistician with a BSc and MSc in Statistics from the University of Brasília (UnB). My research centres on statistical inference, uncertainty quantification, and simulation, developed through Bayesian and frequentist methods, hierarchical and spatial modelling, and computational tools including R, Python, and Stan. During my MSc, I applied these methods to problems in extreme value theory and dependence structures, working with physical and environmental data, including NASA solar irradiance measurements, alongside separate applied work with large administrative and public-sector datasets. Across these projects, my consistent interest has been how statistical modelling, computation, and uncertainty quantification can be combined to draw reliable conclusions from complex, real-world data.

EDUCATION

Feb 2024	MSc in Statistics	
Jul 2026	Graduate Program in Statistics, University of Brasília (UnB)	Brasília, Brazil
	THESIS	<i>“Statistical Modelling of Spatial Extremes in Complex Systems”</i> Developed a Bayesian hierarchical geostatistical model for spatial extremes with BGEV marginals, allowing all distributional parameters to vary spatially through latent Gaussian structures. The model is designed to capture bimodality, tail behaviour, and spatial heterogeneity within a unified probabilistic framework.
	METHODS	Bayesian hierarchical modelling; Extreme Value Theory; latent Gaussian processes with spatial covariance; support-preserving parameter transformations; prior specification and regularisation; non-centred parameterisation; HMC/NUTS inference in Stan; spatial confounding control via QR projection; MCMC simulation and diagnostics; parameter recovery; uncertainty quantification; and estimation of extreme quantiles.
	RESEARCH	Alongside the dissertation, developed independent research in statistical methodology, extreme value theory, climatology, and astrostatistics, resulting in 4 submitted manuscripts. 2 further manuscripts are being developed from the thesis itself, one methodological and one applied. Additional research conducted during the MSc led to 3 oral/invited presentations and 6 scientific posters at national and international events.
	COURSES	Statistical Inference, Bayesian Inference, Mathematical Statistics, Computational Techniques in Statistics, Machine Learning, Geostatistics, Extreme Value Theory and Copulas, Generalized Linear Models.
	STATUS	Thesis defended. All academic requirements completed. Formal diploma issuance pending.
Aug 2018	BSc in Statistics	
Dec 2023	University of Brasília (UnB)	Brasília, Brazil
	THESIS	<i>“Imputation Methods for Missing Data Using Predictive Modelling”</i> Monte Carlo simulation study comparing single and multiple imputation strategies, analysing regression and machine learning methods across different missing data scenarios.
	TRAINING	Probability and statistical inference; regression; Bayesian methods; computational statistics; stochastic processes; time series; multivariate analysis; computing; experimental design; calculus; numerical methods; linear algebra; fundamentals of physics; fundamentals of astronomy and astrophysics.
	RESEARCH	CNPq-funded undergraduate research project on <i>“Imputation Methods for Dichotomous Variables”</i> . Research on simple and multiple imputation methods through simulation and real data analysis in R.
Sep 2022	Academic Exchange on Modelling and Data Science	
Mar 2023	Lodz University of Technology	Łódź, Poland
	On campus coursework in statistical physics, experimental physics, stochastic processes, computational modelling, computer science, and logic, completed in partnership with the University of Brasília.	

RESEARCH OUTPUTS

SUBMITTED MANUSCRIPTS AND PREPRINTS

- [1] **Extremes of Solar Spectral Irradiance in the SORCE/XPS Record**
Brasil, E.; Otiniano, C.; Brito, C.; Albernaz, B.; Morales, F.
Submitted to *Frontiers in Astronomy and Space Sciences*, Astrostatistics topic.
Preprint [astro-ph.SR]: [arXiv:2608.30878](#)
- [2] **Multiple Imputation Methods under Extreme Values**
Brasil, E.
Submitted to the *Austrian Journal of Statistics*.
Preprint [stat.CO]: [arXiv:2602.04751](#)
- [3] **Joint return levels of maximum temperature and minimum relative humidity by combining copulas with an extreme value framework for bimodal data**
Albernaz, B.; Otiniano, C.; Brito, C.; Morales, F.; Brasil, E.
Submitted to *Advances in Statistical Climatology, Meteorology and Oceanography*.
Preprint [stat.ME]: [arXiv:2608.10252](#)
- [4] **Conditional copula representations and extremal bounds for multivariate statistical functionals**
Vila, R.; Otiniano, C.; Brito, C.; Brasil, E.
Submitted to *Statistics and Probability Letters*.
Preprint [math.ST]: [arXiv:2607.26256](#)

CORE METHODS, SCIENTIFIC COMPUTING & DATA

Statistical inference, uncertainty & hierarchical modelling
Bayesian and frequentist inference; hierarchical and latent-variable models; Gaussian processes; parameter inference; uncertainty quantification; model criticism and diagnostics.

Simulation, calibration & model assessment
Monte Carlo simulation; simulation studies; synthetic-data experiments; parameter recovery; calibration and sensitivity analysis; posterior and MCMC diagnostics; robustness assessment.

Dependence, extremes & complex stochastic structure
Extreme Value Theory; GEV/BGEV models; copulas; multivariate and extremal dependence; spatial and spatiotemporal modelling; covariance structures.

Scientific & Bayesian computing
R/RStudio/R Markdown for statistical modelling and simulation; Python for scientific computing, data analysis, and automation; Stan for hierarchical Bayesian inference with HMC/NUTS.

Astrostatistical & physical data analysis
astrostatistical applications; NASA SORCE solar spectral irradiance; space science and time indexed physical measurements; heterogeneous observational datasets; data validation; uncertainty assessment.

Quantitative social & public-sector research
Education indicators, national microdata and longitudinal records; public policy; diversity, equity and inclusion; administrative data validation.

Climate & atmospheric systems
Solar irradiance and air density; temperature and humidity extremes; climate and financial risk.

Reproducible data workflows
SQL, DBeaver, ODBC; LaTeX; Git/GitHub; reproducible reporting; Shiny and Power BI; Stata and SAS.

RESEARCH & PROFESSIONAL EXPERIENCE

Sep 2025 Present	Statistician HU Brasil, Federal University Hospitals	Brazil
	• Analysis and creation of large observational and administrative datasets within the Brazilian public sector, with emphasis on data quality, validation, robustness, and reproducible research. • Develop automated analytical pipelines, dashboards, and research outputs; contribute to published and ongoing studies on diversity, equity and inclusion, social impacts, and psychological research.	
May 2025 May 2026	Senior Research Assistant and Data Science Fellow Ipea, Institute for Applied Economic Research	Brazil
	• Conducted qualitative and quantitative research with large national microdata, longitudinal datasets, and territorial information, integrating and validating heterogeneous sources across spatial and temporal scales. • Developed reproducible statistical workflows, automated analyses, and dashboards; coauthored three technical research notes and a forthcoming book chapter on Brazilian education, with findings communicated through scientific events.	
Aug 2024 Dec 2025	Technical Officer, International Cooperation Mercosul Inep, National Institute for Educational Studies and Research	Brazil
	• Supported South American academic cooperation and undergraduate programme evaluation within the Mercosul accreditation system, including liaison with Latin American partners in meetings conducted in Portuguese and Spanish.	
Sep 2023 May 2025	Data Analyst and Consultant Alcance Profissional	Brazil
	• Built automated data workflows, reproducible analytical pipelines, and reporting processes, with focus on data organization, efficiency, and repeatability.	
Oct 2021 Sep 2022	Statistics Intern FGCoop, Brazilian Credit Cooperative Guarantee Fund	Brazil
	• Supported statistical monitoring, indicator analysis, risk modelling activities, and quantitative reporting.	

FELLOWSHIPS, SCHOLARSHIPS & DISTINCTIONS

NSF EPSCoR International Research Travel Support
International research support. Supported by NSF EPSCoR and the Center for Space Plasma and Aeronomic Research (CSPAR), The University of Alabama in Huntsville, for an oral astrostatistics presentation at UAH and subsequent publication support for the associated *Frontiers in Astronomy and Space Sciences* manuscript.

CNPq MSc fellowship and Admission Distinction
Ranked 1st in MSc admission. First among 20 admitted candidates to the MSc in Statistics at UnB; awarded a two-year CNPq graduate scholarship.

Ipea National Research Fellowship
Ranked 1st for the project. Selected for Ipea’s data science research team through a nationwide competitive selection (12 months).

CNPq and UnB Undergraduate Research Support
Undergraduate research funding. Institutional Scientific Initiation Scholarship Programme and academic teaching assistantship support (18 months).

PPGEST/UnB Scientific Travel Funding
Competitive scientific travel support. Awarded for research presentations at national and international scientific events from 2024 to 2026.

SCIENTIFIC PRESENTATIONS

Jun 2026 Invited speaker	Extremes in Nature: statistics, uncertainty, and forecasting in the face of extreme phenomena 1st Workshop of the Graduate Program in Statistics, University of Brasília, Brazil
Nov 2025 Oral	Joint extremes of solar irradiance and air density: an EVT and copula approach to tail uncertainty Systematic and Measurement Errors across the Sciences: Astrostatistics and Data Science, The University of Alabama in Huntsville, USA; astrophysics applications track
Oct 2025 Poster	Analysis of extreme solar irradiance in NASA’s Solar Radiation and Climate Experiment (SORCE) mission: an Extreme Value Theory approach 5th Paraíba Meeting of Statistics, João Pessoa, Brazil
Oct 2025 Poster	Impact of extreme values on multiple imputation of missing data using MICE in R 19th School of Regression Models, João Pessoa, Brazil
Aug 2025 Poster	Climate-financial risk modelling for the agricultural sector using bivariate copulas 69th Brazilian Statistics Meeting (RBras) and 21st SEAGRO, Vitória, Brazil
Aug 2025 Poster	Bivariate analysis of maximum temperature and minimum relative humidity in Brasília (DF) 69th Brazilian Statistics Meeting (RBras) and 21st SEAGRO, Vitória, Brazil
Aug 2025 Poster	Extreme solar irradiance and its relationship with atmospheric variables: an approach using Extreme Value Theory and copulas 69th Brazilian Statistics Meeting (RBras) and 21st SEAGRO, Vitória, Brazil
Sep 2024 Oral	Assessing imputation methods for missing data using statistical models: a simulation approach 1st Brazil and Colombia Binational Colloquium on Probability and Statistics, Natal, Brazil
Sep 2022 Oral	Imputation techniques for dichotomous variables University of Brasília Undergraduate Research Congress, Brasília, Brazil

- 2026 **A Bayesian hierarchical model for spatial extremes with bimodality**
Accepted Poster Brazilian Symposium on Probability and Statistics (SINAPE)
- 2026 **Listening to the workforce: statistical planning of the Diversity, Equity, and Inclusion Census in Brazil's largest public network of university hospitals**
Accepted Poster Brazilian Symposium on Probability and Statistics (SINAPE)
- 2026 **The expansion of full-time education in Brazil (2007–2024): federal programmes, funding, and patterns of enrolment growth**
Accepted Oral 2nd Pan-American Meeting on the Economics of Education
- 2026 **Full-time education in Brazil: funding, expansion, and challenges under the Fundeb framework**
Accepted Oral 50th Annual Meeting of ANPOCS

ADDITIONAL RESEARCH OUTPUTS

MANUSCRIPTS IN PREPARATION

- [5] **Characterising complex dependence between extreme solar irradiance and air density**
First-author manuscript in preparation; submission planned for Nov. 2026.
- [6] **Bayesian hierarchical modelling of spatial extremes with BGEV marginals**
First-author manuscript in preparation; submission planned for Jan. 2027.

INTERDISCIPLINARY CONTRIBUTIONS

- [7] **Labor Relations at the Brazilian Hospital Services Company: Democratization and the Promotion of Diversity, Equity, and Inclusion**
Dau, D.; Junqueira, E.; Brasil, E.; Lobo, L.; Viana, L.; Maia, M.; Sousa, T.
Revista Jurídica da Empresa Brasileira de Serviços Hospitalares, 2(2), 2025; published in Portuguese.
- [8] **Full-Time Education Dashboard: Data Sources, Implementation, and Analytical Use**
Brasil, E.; Resende, M.; Nascimento, P. M.
Technical research note, Ipea, 2026; in Portuguese.
Editorial review.
- [9] **Calculation and Presentation of Educational Indicators in the 2024–2027 Multi-Year Plan**
Brasil, E.; Candeia, L.; Rocha, V.; Senkevics, A.
Technical research note, Ipea, 2026; in Portuguese.
Editorial review.
- [10] **The Evolution of Full-Time Education in Brazil**
Resende, M.; Santos, F.; Brasil, E.; Brito, A.
Technical research note, Ipea, 2026; in Portuguese.
Editorial review.
- [11] **Full-Time Education in Brazil: Financing, Expansion, and Challenges in Light of Fundeb**
Xavier, A.; Senkevics, A.; Brasil, E.; Rocha, V.
Book chapter, 2026; in Portuguese.
Forthcoming.

SELECTED ADVANCED TRAINING

Statistical inference, simulation & scientific computing

- 1st Workshop on Neural Simulation-Based Inference *The Statistical Methods for the Physical Sciences, Carnegie Mellon University* (Oct 2025)
- Statistical Modelling of Digital Images *5th Paraíba Meeting of Statistics* (Oct 2025)
- R and Python with Spark for big-data *5th Paraíba Meeting of Statistics* (Oct 2025)
- Missing Data in Longitudinal Analysis *25th SINAPE, Fortaleza* (Aug 2024)
- Regression Models for Heavy-Tailed and Asymmetric Data *17th School of Regression Models, IBGE* (Dec 2021)
- Machine Learning with `tidymodels` in R *17th School of Regression Models, IBGE* (Dec 2021)

Astronomy, astrophysics & physical sciences

- Astronomy *Brazilian Space Agency* (May 2026)
- Introduction to Physical Oceanography *Brazilian Space Agency* (Mar 2026)
- Quantum and Statistical Physics Workshop *International Center of Physics, University of Brasília* (Apr 2024)
- Fundamentals of Astronomy and Astrophysics *University of Brasília* (2020)
- AstroTech: Science and Technology behind Astronomical Discovery *University of Edinburgh* (Jul 2020)
- Quantum Mechanics *École Polytechnique* (Jul 2020)
- Origins of Life in the Cosmic Context *University of São Paulo* (Jul 2020)

Research practice & reproducibility

- Research Data Management, Collaborations, and Integrity *Elsevier and Springer Nature* (Aug 2025)
- Climate Change and Sovereign Risk *Asian Development Bank Institute* (Feb 2025)
- R, Python, R Markdown, Data Cleaning, and Tidyverse Workflows *DataCamp* (Aug–Sep 2021)
- Mathematical Foundations for Cryptography *University of Colorado* (Jul 2020)

ACADEMIC SERVICE, TEACHING & VOLUNTEERING

Academic leadership & service

- Jun 2026 **Organizing committee member** 1st Extension Forum, Institute of Exact Sciences, University of Brasília
- 2025 **Peer reviewer (6 manuscripts)** *Computational Statistics*, Springer Nature
- 2024 **Academic event monitor** 7th Workshop on Survival Analysis and Applications, University of Brasília
- Jan 2022 **General coordinator** 50th Summer School and 14th Summer Workshop in Mathematics, University of Brasília
- May 2021 **Academic event monitor** 6th Colloquium on Mathematics of the Central-West Region, University of Brasília

Teaching & academic support

- Mar–Jul 2025 **Graduate Teaching Internship in Statistics** University of Brasília
- 2019 **Teaching Assistantship in Probability Theory** University of Brasília
- Long-term **Academic Tutoring and Mentoring** Secondary school and undergraduate students on statistics, mathematics, and physics

LANGUAGES

Portuguese: native. **English:** fluent. **French:** intermediate. **Spanish:** reading and listening comprehension. **Polish:** introductory exposure.